Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: medium

Benchmark: *LTZGLUE* | ***Context:*** *Luxembourgish Professional Newsroom*

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Code-switching is identified as a HIGH-priority deployment requirement [A2] and is a structural feature of Luxembourgish journalism [WEB-6, WEB-10]. The benchmark explicitly acknowledges that 'most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts' [Q112], and that data are concentrated in a limited set of public domains [Q119]. Only the human-annotated RTL news comment NER corpus is documented as covering 'informal and code-mixed writing' [Q41]; systematic quantification of code-switched content across the other tasks (HA, LA, TC) is absent. RTL is the principal source for HA, SA, TC, and NER comments [Q16, Q24, Q40, Q46], which aligns with the deployment's primary outlet [WEB-1], a meaningful strength. However, LA sentences are sourced from the LOD formal dictionary [Q30], RTE is machine-translated then LLM-improved [Q60, Q61], and intent detection uses German MT [Q54] — meaning much of the benchmark content is not authentic Luxembourgish journalistic text. Given code-switching is a HIGH-priority dimension for this deployment and the benchmark itself flags this as a limitation, score is 2.

ID	Evidence
Q16	“To construct this dataset, we use RTL news articles. We keep only documents from the twenty most frequent categories. We then filter articles by body length and title length, remove exact duplicate titles, randomly shuffle the remaining instances, and retain a fixed subset of 30k examples.” (p.3)
Q24	“We use articles from RTL, randomly selected from the commentary and letter to the editor sections.” (p.3)
Q30	“The sentences are derived from the Luxembourgish Online Dictionary (LOD) and are manipulated using the tags available in the dataset.” (p.3)
Q40	“The NER dataset introduced by Lothritz et al. (2022), by contrast, is a fully human-annotated corpus derived from RTL online news comments.” (p.4)
Q41	“It covers a wider range of text types and registers, including informal and code-mixed writing, and focuses on four primary entity categories (PER, ORG, LOC, GPE).” (p.4)
Q46	“To construct the news topic classification (TC) dataset, we collected news articles from RTL, which provides content pre-assigned to editorial categories.” (p.4)
Q54	“Since this task is originally intended to be crosslingual, we use the machine translated German training set (van der Goot et al., 2021).” (p.5)
Q60	“Lothritz et al. (2023) released a machine-translated Luxembourgish version of the dataset using Google Translate. However, due to numerous grammar and vocabulary related mistakes introduced in this process, we set out to improve the quality of the dataset.” (p.5)
Q61	“Specifically, we first prompted CHATGPT-5.1 to assess and improve the translated sentence pairs unless they were already of very high quality, while explicitly keeping the original meaning to avoid label conflicts (see Appendix 7.4).” (p.5)
Q112	“Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts.” (p.9)
Q119	“LTZ is a small language community, and linguistic data often originate from a limited set of public domains.” (p.10)
WEB-1	Luxembourgish news routinely covers cross-border (Belgian, French, German) entities, motivating GPE/LOC granularity (https://media-ownership.eu/2023-edition/findings/countries/luxembourg/)
WEB-6	Luxembourg's three-language administrative regime (Law of 24 February 1984) institutionalises French/German/Luxembourgish co-occurrence in journalistic text (https://en.wikipedia.org/wiki/Languages_of_Luxembourg)

WEB-10	Luxembourgish nationals speak on average 4.3 languages and 9 of 10 residents speak French, confirming code-switching as a structural feature (https://en.paperjam.lu/article/delano_23-lux-residents-speak-4-or-more-languages)

Strengths

- RTL — the principal Luxembourgish broadcast outlet [WEB-1] — is the dominant source for HA, SA, TC, and NER, providing strong domain alignment for the deployment's primary register [Q16, Q24, Q40, Q46]
- The RTL news comment NER subcorpus explicitly covers informal and code-mixed writing [Q41], partially addressing the deployment's code-switching need for NER
- The benchmark explicitly acknowledges its register and code-switching limitations [Q112, Q119], enabling informed deployment risk management

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Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — Luxembourgish journalism requires handling French, German, and English code-switching, EU institutional terminology, and Grande Région cross-border entities [A2, WEB-1, WEB-10]. The benchmark partially addresses this through the RTL comment NER subcorpus [Q41] but does not systematically cover code-switching across HA, LA, or TC.

IC-2: RTL-derived content (HA, SA, TC, NER comments) [Q16, Q24, Q40, Q46] aligns with mainstream Luxembourgish journalistic culture; LA sentences from LOD [Q30] reflect formal/normative register diverging from the deployment's flexible journalistic standard [A3].

IC-3: Intent detection uses xSID voice-assistant commands [Q51, Q55] — a register entirely irrelevant to newsroom deployment. RTE is sourced from English-origin pairs translated and LLM-improved [Q60, Q61], introducing non-Luxembourgish provenance.

IC-4: INSUFFICIENT DOCUMENTATION — no regional annotator audit of code-switched coverage across the benchmark has been performed; gap_id 1 in the regional YAML remains unresolved [NEEDS VERIFICATION]. Direct dataset inspection by Luxembourgish journalism experts would be needed.

IC-5: Documented content issues: (a) systematic under-representation of code-switching outside the NER comment subcorpus [Q112]; (b) LA grounded in formal LOD register [Q30] rather than journalistic register; (c) 22–28% of RTE filtered post-LLM improvement [Q66] indicating quality variance; (d) JUDGEWEL automatically constructed with minimal human verification [Q37].

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Q24	"We use articles from RTL, randomly selected from the commentary and letter to the editor sections." (p.3)
Q30	"The sentences are derived from the Luxembourgish Online Dictionary (LOD) and are manipulated using the tags available in the dataset." (p.3)
Q37	"The resulting sentences are then evaluated using LLMs acting as judges, with minimal human verification to calibrate quality thresholds." (p.4)
Q40	"The NER dataset introduced by Lothritz et al. (2022), by contrast, is a fully human-annotated corpus derived from RTL online news comments." (p.4)
Q41	"It covers a wider range of text types and registers, including informal and code-mixed writing, and focuses on four primary entity categories (PER, ORG, LOC, GPE)." (p.4)
Q46	"To construct the news topic classification (TC) dataset, we collected news articles from RTL, which provides content pre-assigned to editorial categories." (p.4)
Q51	"We constructed a new LTZ dataset for intent detection (ID) by translating the English xSID test and validation datasets (van der Goot et al., 2021)." (p.4)
Q55	"The main challenge in translating the English dataset stems from its register. The source segments consist of user commands for a voice-controlled AI assistant, representing a specialised spoken register for which there is no equivalent reference corpus in LTZ. This register is marked by domain-specific terminology and collocations (e.g., set an alarm, set a reminder, add to playlist), as well as non-standard spelling (e.g., all lowercase, missing punctuation)." (p.5)
Q60	"Lothritz et al. (2023) released a machine-translated Luxembourgish version of the dataset using Google Translate. However, due to numerous grammar and vocabulary related mistakes introduced in this process, we set out to improve the quality of the dataset." (p.5)
Q61	"Specifically, we first prompted CHATGPT-5.1 to assess and improve the translated sentence pairs unless they were already of very high quality, while explicitly keeping the original meaning to avoid label conflicts (see Appendix 7.4)." (p.5)
Q66	"The filtering reduced between 22 and 28% of instances in the data, resulting in a final dataset of 1,876, 197, and 626 sentence pairs for the training, development, and test set, respectively." (p.5)
Q112	"Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts." (p.9)
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Information Gaps

- Quantified rate of French/German/English code-switching in HA, LA, and TC instances (gap_id 1, deferred)
- Whether RTL article corpus reflects only published staff articles or includes user-comment register
- Coverage of EU institutional naming conventions and Grande Région entity mentions in the NER training set

Requires Expert Verification

- Whether code-switching density in benchmark instances matches operational density in target newsroom copy
- Whether the RTL-sourced subset reflects the same editorial register as the deploying outlet's output

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Remediation

Gap 1: Code-switching density across HA, LA, and TC tasks is undocumented; only the NER comment subcorpus explicitly covers code-mixed text [Q41, Q112]

Recommendation: Sample 200–500 instances per task and label code-switched tokens; commission a code-switched evaluation supplement drawn from RTL articles to quantify performance under realistic French/German/English mixing

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